

kea: A Python package for consistent power spectral analysis of data on arbitrary dimensions

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Summary

Power spectral density techniques are widely used across disciplines but often differ in normalization and data handling, with most fields relying on a limited set of established methods. Historically, researchers have been required to implement their own routines or personally request code from others. This increases barriers to entry for performing analysis and requires a deep understanding of specific Fourier conventions to avoid errors. Without a standardized framework, order-unity normalization difference often creep in, which can lead to physical inferences that are mathematically inconsistent or physically erroneous. To address this, we present a computational package that provides consistently normalized estimators that are adaptable to multiple data types – including intracluster medium surface brightness fluctuations, in-situ solar wind time-series, and simulation cubes.

Statement of need & state of the field

Autocorrelation functions (ACF), power spectral densities (PSD), and structure functions (SF) serve as the primary mathematical framework for quantifying the statistical properties of stochastic phenomena across varying spatial and temporal scales. In the context of fluid dynamics and astrophysics, these tools allow researchers to decompose complex, multiscale signals into their constituent parts. Of particular interest, especially within the study of (magneto)-hydrodynamic turbulence, are the characteristic scales (such as the integral scale L where energy is injected, and the dissipation scale η), the power-law indices that define the energy cascade (e.g., the Kolmogorov $-5/3$), and their respective amplitudes which dictate the total turbulent energy budget.

These phenomena are rarely captured in the full continuum of values in the available space in which measurements occur. In other words, in a lot of practical cases, observations are restricted to only 1D slices, 2D projections, or 2D slices. For example, in the case of the solar wind, in-situ measurements are performed by sensors moving in relation to the plasma rest frame. These sensors generate a time-series of data: such as magnetic field (vector field), or density (scalar field) measurements. As another pertinent example, surface brightness observations of the intracluster medium (ICM) observe 2D (emission weighted) projections of scalar fields, or projections of line-of-sight components of a vector field. Often these observations are calibrated or compared against more accessible regimes, or high-fidelity simulations.

Beyond the choice of statistical tool, the reliability of the output is heavily dependent on the nuances of spectral estimation. There exists a vast library of signal processing techniques designed to mitigate the distortions, aliasing, and biases inherent in finite sampling ([Maciejewski](#)

et al., 2009; Sefusatti et al., 2016; Stoica & Moses, 2005). However, the implementation of these techniques introduces its own set of variables: how the data are windowed to prevent spectral leakage, how the estimates are binned in k -space, and how the resulting power is normalized (e.g., ensuring Parseval's theorem is satisfied). For example, there are a total of six different R packages that provide spectrum estimations with varying normalization options(Barbour & Parker, 2022).

If these methodological choices are not standardized, they can induce spurious physical behavior. For example, improper binning can artificially flatten a spectral slope. This could become a significant issue when comparing across different studies or when cross-correlating observational data with numerical models. Without a rigorous, consistent approach to these estimations, the resulting physical interpretations (such as the injection or dissipation scales) may reflect the limitations of the signal processing rather than the underlying physics.

While the mathematical definitions of autocorrelation functions, power spectral densities and structure functions are well-established, their application to "real-world" datasets show significant variability in subtle ways. For example, normalizations of the difference-of-Gaussian method differ (Arévalo et al., 2012; Churazov et al., 2012; Zhou et al., 2022).

Currently available software for ICM turbulence analysis are: turbustat (Koch et al., 2019), and PITSZI (Adam et al., 2025). Both of these solutions require 2D data, thus are not available for applications to e.g., solar wind turbulence analysis, or simulation cubes. Additionally, they both use a nested sequence of object-oriented class based design with built-in handling with assumptions for their respective fields: the interstellar medium for turbustat, and the intracluster medium for PITSZI. Hence, modifying for individual needs is challenging. SF calculation is available using fastSF (Sadhukhan et al., 2021), however, only for uniform 2D and 3D datasets. In other words, fastSF is not applicable to gapped datasets. SF calculation is also available in Python with fluidsfs (Wagner et al., 2025).

We thereby introduce Kea (Analysis Engine MarK, backwards), a Python package that implements several different dimensionally-agnostic PSD estimation techniques for use in turbulence analysis. Kea is available on GitHub via <https://github.com/mab68/Kea>.

Implementation Details

Synthesis of Stochastic Fields

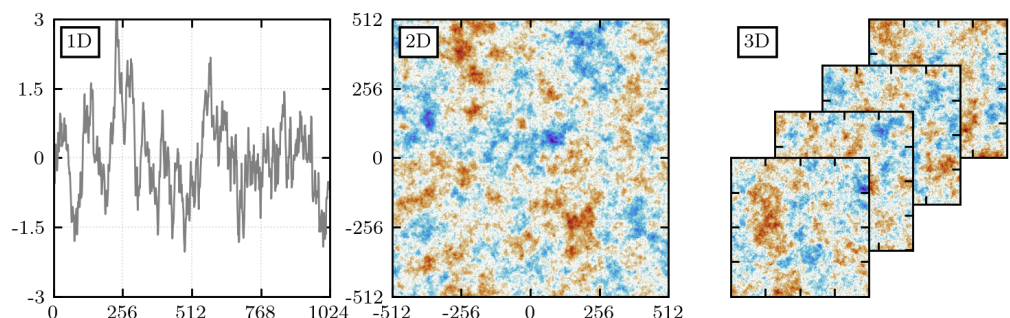


Figure 1: Synthetic monofractal test fields generated by Kea, demonstrating the package's support for 1D (left), 2D (middle), and 3D (right, shown as a 2D slices) spatial data structures. These generation tools allow users to easily create controlled datasets for testing and benchmarking.

Power-law correlations are a key feature of turbulence (particularly with high Reynolds numbers flows). The scale-to-scale energy transfer associated with the cascade of energy from large

scales to smaller scales leads to a power-law power spectrum which is indicative of fractal-like structure where small-scale features are statistically similar to large-scale ones.

Observing the complete set of information provided by self-similar physical phenomena can be practically challenging. In particular, we highlight that the majority of astronomy and astrophysical observations are driven by the detection and manipulation of photons. The properties of these photons are determined by the conditions of the source (temperature, density, velocity, etc.) of the emitting material and its subsequent interaction with any intervening material. Determining the conditions of the source from the photons alone can be a serious challenge. Noise and resolution constraints may also limit the accuracy of the inference and increase complexity. To understand biases and systematics or comparison of theory to observation, where data is lacking, it can be necessary to compute expected observational properties from controlled surrogate models (Haworth et al., 2018; Simionescu et al., 2019).

For example, synthetic fields have been used extensively to validate methods and measure turbulence in the interstellar medium (Brunt & Heyer, 2002; Esquivel et al., 2003; Miville-Deschênes et al., 2003; Ossenkopf et al., 2006) and similarly for the ICM (Vogt & EnBlin, 2005; XRISM Collaboration et al., 2025; ZuHone et al., 2016).

kea provides methods to produce (monofractal and multifractal) synthetic fields with correlations parameterized by power spectral density. These fields are synthesized in Fourier-space and Fourier-transformed to provide real-space fluctuation fields (Barnsley et al., 1988; Lakhali et al., 2025). For examples, shown in Figure 1 are one, two, and three-dimensional monofractal fields. Thus, the below PSD estimators can be tested for validity with known/expected forms, and their performance tested under constraints of noise and missing data.

Fourier Convention

The data we obtain from real observations is not continuous data, but rather, a discrete function that is obtained by sampling an underlying continuous function, y_c , at an interval Δx . This gives the discrete function $y[n]$ and its Fourier transform $\hat{y}[m]$, defined as the following:

$$y[n] = y_c(n\Delta x) = y_c(x),$$

$$\hat{y}[m] = \hat{y}_c(m\Delta k) = \hat{y}_c(k),$$

where $x = n\Delta x$, and $k = m\Delta k$ with $m, n \in \mathcal{Z}$ (integers).

We assume that $y_c(x)$ is periodic on the domain $x \in [0, L]$ such that $y_c(x) = y_c(x + L)$ and we have sampled N evenly separated points within this domain. Our sampling interval is therefore $\Delta x = L/N$. Here, L represents the physical domain and N the number of grid points. The discrete function will also be periodic: $y[n] = y[n + N]$ for $n \in [0, N - 1]$, $m \in [-N/2, N/2]$. As a result, the Fourier-space function is also periodic: $\hat{y}_c(k) = \hat{y}_c(k + N\Delta k)$ where $\Delta k = \frac{2\pi}{N\Delta x} = \frac{2\pi}{L}$ and $k \in [-\pi N/L, \pi N/L]$. Alternatively, a non-periodic signal is assumed to be $y_c(x) = 0$ for $x > L$. Note that mathematically, the DFT treats the signal as periodic regardless so the function should be padded with zeros to ensure there is minimal spectral leakage due to a discontinuity at the boundary.

The last step in this process is to acknowledge the following relation for the discretization of the complex exponential term: $xk = (n\Delta x)(m\Delta k) = nm\Delta x 2\pi/(N\Delta x) = 2\pi nm/N$. Applying the relations stated gives us the discrete form of the continuous signal and its Fourier transform, where we have also replaced dx, dk with $\Delta x, \Delta k$ respectively (Allen et al., 2012):

$$y[n] = \Delta k \sum_{m=-N/2}^{N/2} \hat{y}[m] e^{2\pi i n m / N},$$

$$\hat{y}[m] = \frac{\Delta x}{2\pi} \sum_{n=0}^{N-1} y[n] e^{-2\pi i n m / N}.$$

We can derive the generalized D -dimensional form of the DFT using the same arguments provided above, which we state as the following:

$$y[\mathbf{n}] = (\Delta k)^D \sum_{\mathbf{m}} \hat{y}[\mathbf{m}] e^{2\pi i \mathbf{n} \cdot \mathbf{m} / N},$$

$$\hat{y}[\mathbf{m}] = \left(\frac{\Delta x}{2\pi} \right)^D \sum_{\mathbf{n}} y[\mathbf{n}] e^{-2\pi i \mathbf{n} \cdot \mathbf{m} / N},$$

where the summations are over all $\mathbf{m}, \mathbf{n}$ e.g., $\sum_{\mathbf{n}} = \sum_{n_1=0}^{N_1-1} \sum_{n_2=0}^{N_2-1} \dots \sum_{n_D=0}^{N_D-1}$.

The D -dimensional DFT is implemented using the NumPy.fft routines which is then multiplied by the correct scaling factor. The DFT is called using the NumPy.fft.fftn method, and the wavenumbers are obtained using the NumPy.fft.fftfreq, providing the number of sampled data points N (this gives m/N for $m \in [-N/2, N/2]$) which is subsequently multiplied by $2\pi/\Delta x$ to get the *angular* wavenumbers.

The DFT is well-defined when $y[\mathbf{n}] \in \mathcal{R}$, however, particularly it is not always the case that $y[\mathbf{n}]$ has such values for all $\mathbf{n} \in \{1, \dots, N-1\}$ with constant $\Delta x = L/N$. This can be due to a myriad of reasons depending on the observation type, but generally, it will result in “gaps” in the data. To retain uniform sampling, a gap in $y[\mathbf{n}]$ is represented by NaN (not a number; NumPy.nan). However, most implementations of the above discrete Fourier transforms do not behave when $y[\mathbf{n}]$ has NaN values. Hence, alternative spectral estimation methods are frequently used.

Lag Functions

We discuss the class of functions termed the “lag-function” which use the same backend i.e., performing some operation (where the ACF performs multiplication, and the SF performs subtraction) on data that has been shifted by a lag distance.

Autocorrelation Function

The autocorrelation function is estimated via:

$$R[\mathbf{m}] = \frac{1}{\tilde{N}[\mathbf{m}]} \sum_{\mathbf{n}} y[\mathbf{n}] y[\mathbf{n} + \mathbf{m}], \quad (1)$$

where $\tilde{N}$ is a normalization by the number of data points, and $\mathbf{m}$ is a lag-distance in real-space data (rather than the wavenumbers, above), and the summation is over the *available* data (i.e., we ignore NaN values). The shift has to cut both $y[\mathbf{n}]$ and $y[\mathbf{m}]$ such that only the overlapping sequence (of equal length $N - m$) is present (Sadhukhan et al., 2021).

There are typically two standard ways of normalizing Equation 1, either: $\tilde{N}[m] = N - m - 1$, or $\tilde{N}^{\text{biased}}[m] = N$. For D -dimensions, $\tilde{N}[\mathbf{m}] = (N_1 - m_1 - 1) \dots (N_D - m_D - 1)$ and $\tilde{N}^{\text{biased}}[\mathbf{m}] = N^D$. If $\tilde{N}[m] = N - m - 1$ is used, then Equation 1 is sometimes called the standard unbiased ACF estimate. In this case, for large lags the factor $\frac{1}{\tilde{N}[m]}$ becomes large; because the average is calculated over relatively few products, the estimate is not statistically well-constrained and is prone to high variance. However, in practice large m samples inhomogeneities beyond the correlation scale, so only up to a maximum lag of $m_{\text{max}} = \frac{N}{2}$ (or even $m_{\text{max}} = \frac{N}{4}$) should be taken. In which case, the unbiased estimator (with $\tilde{N}[m] = N - m - 1$) has $\approx 50\% - 75\%$ of N at $m = m_{\text{max}}$ and can be statistically reliable. If the biased estimator is used, then Equation 1 more closely resembles the continuous ACF in the case of non-periodic $y[n]$:

$$R[\mathbf{m}] = \left(\frac{\Delta x}{L} \right)^D \sum_{\mathbf{n}} y[\mathbf{n}] y[\mathbf{n} + \mathbf{m}] = \frac{1}{N^D} \sum_{\mathbf{n}} y[\mathbf{n}] y[\mathbf{n} + \mathbf{m}],$$

for $\Delta x = L/N$.

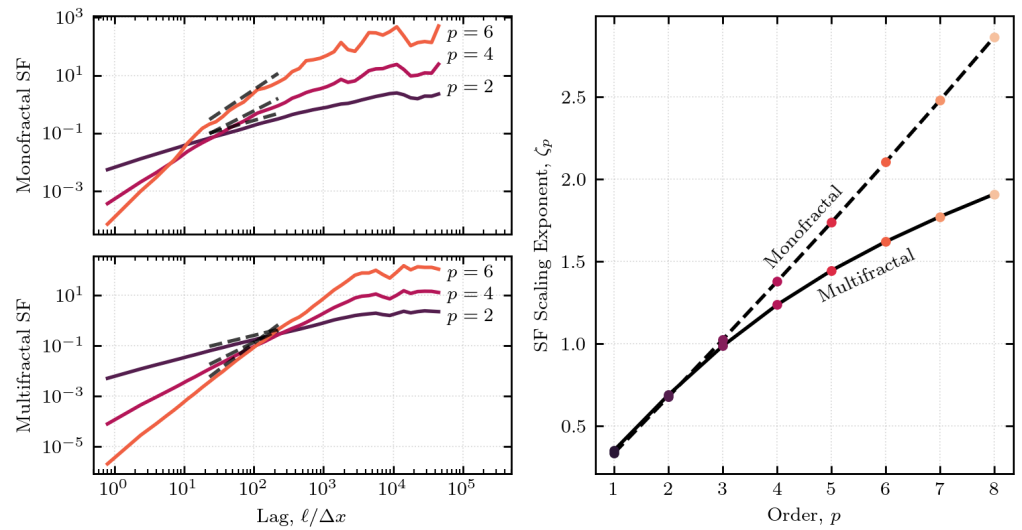


Figure 2: Capabilities of Kea for structure function scaling analysis. The package is used to generate 1D synthetic monofractal and multifractal fields, calculate structure functions from order $p = 1$ to $p = 8$, and extract the corresponding power-law slopes.

154 The SF of order p follows the ACF, as,

$$S^{(p)}[\mathbf{m}] = \frac{1}{\tilde{N}[\mathbf{m}]} \sum_{\mathbf{n}} |y[\mathbf{n}] - y[\mathbf{n} + \mathbf{m}]|^p,$$

155 for the choice of $\tilde{N}[\mathbf{m}] = (N_1 - m_1 - 1) \dots (N_D - m_D - 1)$, and we ignore NaN values. The
156 unbiased estimator is typically used with (at most) $m_{\max} = \frac{N}{2}$.

157 Kea also facilitates the calculation of higher-order scale-dependent statistics using the order
158 parameter p . This can be used to e.g., evaluate mono- and multi-fractal scaling behaviour as
159 shown in Figure 2; an approach fundamental to modern turbulence studies (Biskamp, 2003;
160 Frisch, 1995).

161 Spectral Estimates

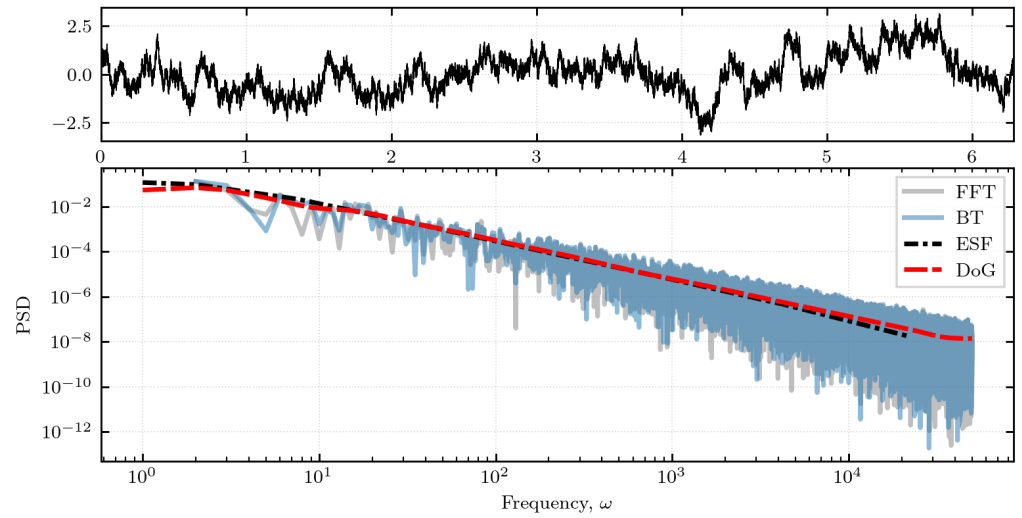


Figure 3: Spectral estimation capabilities of Kea. The top panel displays a 1D synthetic timeseries generated by the software, while the bottom panel illustrates the resulting spectral estimates computed using Kea's implementations of the periodogram (FFT), Blackman-Tukey (BT), equivalent spectrum (ESF), and difference-of-Gaussian (DoG) methods.

162 Kea provides discrete implementations of several spectral estimation techniques. An application
163 of these methods to a synthetic timeseries is demonstrated in Figure 3. We introduce each of
164 the supported methods below.

165 Periodogram

166 The periodogram spectral estimate is simply, (Schuster, 1898)

$$E^{\text{FFT}}[\mathbf{m}] = |\hat{y}[\mathbf{m}]|^2 \Delta k^D.$$

167 This is an estimation of the modal spectrum (hence the vector $\mathbf{m}$ argument) which needs to
168 be subsequently angle-averaged or angle-integrated. When $y[\mathbf{n}]$ has NaN values, often these
169 values are instead interpolated (e.g., with the average, or linearly).

170 Correlogram/Blackman-Tukey

171 The Blackman-Tukey (BT; sometimes called a correlogram) method (Blackman et al., 1960)
172 uses the (discrete) Fourier transformation of the ACF given by,

$$E^{\text{BT}} = \Re \{ \widehat{Q}[\mathbf{m}] \},$$

173 where $\widehat{Q}[\mathbf{n}]$ is the (discrete) Fourier transform of the ACF multiplied by a window function,
174 $W[\mathbf{n}]$:

$$Q[\mathbf{n}] = R[\mathbf{n}]W[\mathbf{n}].$$

175 Note that with the unbiased normalization ($\widetilde{N}[m] = N - m - 1$), $\widehat{Q}[\mathbf{m}]$ is not guaranteed to
176 be positive semi-definite, possible resulting in negative spectral estimates which is undesirable
177 in most applications (Stoica & Moses, 2005).

178 Equivalent Spectrum

179 To estimate a power spectrum using the structure function, first, one must:

- 180 ▪ Calculate the second-order structure function: $S^{(2)}[\mathbf{m}]$.
- 181 ▪ Average (bin) the second-order structure function over shells of magnitude $m = |\mathbf{m}|$:
- 182 $\bar{S}^{(2)}[m]$.

183 Then, the following algorithm is performed (Bishop et al., 2026):

- 184 ▪ Estimate an “uncorrected” equivalent spectrum, $\tilde{\mathcal{E}}^{\text{ESF}}[m]$ using the following relationship:

$$\mathcal{E}^{\text{ESF}}[m] = \frac{1}{2} \frac{1}{b} (m\Delta x)^2 \frac{\Delta \bar{S}^{(2)}[m]}{\Delta(m\Delta x)},$$

185 where $\frac{\Delta \bar{S}^{(2)}[m]}{\Delta(m\Delta x)}$ represents a finite-difference estimate of the lag-space derivative of the
 186 structure function. In the above equation, m still represents the lag-shift. To convert m
 187 to a wavenumber, a factor b is required.

- 188 ▪ Estimate the local power-law slope of the “uncorrected” spectrum.
- 189 ▪ Use an analytical expression for the bias for a pure power-law to derive a wavenumber
 190 dependent correction factor.
- 191 ▪ Return a “debiased” spectrum: $\mathcal{E}^{\text{ESF}}[m]$.

192 kea performs the above algorithm and returns both the “uncorrected” and “debiased” spectral
 193 estimates along with the associated (equivalent) wavenumbers. For the mathematical
 194 justification and validation of this method of a range of datatypes see (Bishop et al., 2026).

195 Difference of Gaussian

196 The difference-of-Gaussian method is described for the continuous case with no gaps; there
 197 are additional convolutions that can be performed to account for gaps in the data (Arévalo et
 198 al., 2012; Ossenkopf et al., 2008).

199 First, we define the real-space scale σ , and the corresponding pixel-space scale o as,

$$\sigma = o\Delta x = bk^{-1} = b(m\Delta k)^{-1}.$$

200 For $k = \{\frac{2\pi}{L}, \dots, \frac{\pi N}{L}\}$, then $\sigma = \{\frac{bL}{\pi N}, \dots, \frac{bL}{2\pi}\}$ and $o = \{\frac{b}{\pi}, \dots, \frac{bN}{2\pi}\}$. Similar to the ESF
 201 method, $b = \pi$ would be a natural conclusion to arrive at based on a signal in a domain L ,
 202 however, it has been suggested otherwise (Bishop et al., 2026). A common interpretation is
 203 $b = \sqrt{2}$ (Arévalo et al., 2012).

204 For the pixel-scale o , the normalized scale-filtered field is,

$$y_o[\mathbf{n}] = M[\mathbf{n}] \Xi[\mathbf{n}] \frac{(y * G_o)[\mathbf{n}]}{(\Xi * G_o)[\mathbf{n}]},$$

205 where `ndimage.gaussian_filter` is used to perform the discrete convolution (by default,
 206 `scipy` is used but if a GPU is configured the `cupyx` library can be used instead). The Gaussian
 207 function is truncated at $10 \times o$ to increase performance (smaller truncations were found to
 208 provide insufficient accuracy in certain circumstances). The function $\Xi[\mathbf{n}]$ represents a (general)
 209 floating-point exposure map (for e.g., X-ray surface brightness observations where $\Xi[\mathbf{n}] = 1$
 210 represents a pixel that has relative complete observation) that also defines the boolean mask,

$$M[\mathbf{n}] = \begin{cases} 1 & \text{where } \Xi[\mathbf{n}] > 0, \\ 0 & \text{otherwise,} \end{cases}$$

211 which ensures we are not counting regions that should be masked. In other words, $\Xi[\mathbf{n}]$
 212 characterizes the locations of the gapped data.

213 For σ values that are nearly equal (with $\xi \approx 10^{-3}$)

$$\sigma_1 = o_1 \Delta x = \frac{\sigma}{\sqrt{1+\xi}}, \sigma_2 = o_2 \Delta x = \sigma \sqrt{1+\xi},$$

214 the scale-filtered variance is,

$$V[o] = \sum_{\mathbf{n}} \left(y_{o_1}[\mathbf{n}] - y_{o_2}[\mathbf{n}] \right)^2,$$

215 which is then normalized to get the angle-averaged spectrum:

$$\bar{E}^{\text{DoG}}[o] = \frac{N^D}{\sum_{\mathbf{n}} \Xi[\mathbf{n}] \sum_{\mathbf{n}} \left(G_{o_1}[\mathbf{n}] - G_{o_2}[\mathbf{n}] \right)^2} V[o] N^{-2D},$$

216 where o is related to the Fourier-space $m = \frac{b}{o} \frac{1}{\Delta x \Delta k}$.

217 Binning

218 Since the ACF, SF, and FFT, and BT spectral estimates provide D -dimensional functions, it is
219 useful to bin them to reduce the dimensionality. Kea provides a general binning routine that
220 applies to the spectral estimates as well as the lag-functions.

221 With bin-width $\Delta b = x_+ - x_-$ and $b \in \mathcal{R}$,

$$F[b] = \widetilde{\sum}_{x_- \leq |\mathbf{n} \Delta x| < x_+} f[\mathbf{n}], \quad (2)$$

222 where the following options are provided as parameters for the routine:

- 223 ■ Provide a binning function $\widetilde{\sum}$ which is typically e.g., an average, or a summation, or a
224 standard error to describe the variation within the bin.
- 225 ■ Whether to ignore the range of values that are outside the circular or spherical shell
226 (since the data is represented on a square/cubic grid).
- 227 ■ Defining specific minimum and maximum b : $b_{\min}, b_{\max}$.
- 228 ■ The location of the number that represents the bin e.g., the center of the bin ($b =$
229 $(x_+ + x_-)/2$), or the left most value ($b = x_-$).
- 230 ■ Whether to apply a normalization of the bin width: Δb^{-1} .
- 231 ■ Bin with log-spaced widths. In which case, the bins are uniformly spaced in log-space.
- 232 ■ Specify a fixed specific number of bins, N_{bins} .

233 Following the continuous definitions the ACF and SF are typically represented via averaging
234 circular/spherical shells¹. Whereas, the angle-integrated ($\mathcal{E}[b]$) and angle-averaged ($\bar{E}[b]$)
235 spectra are found when $\widetilde{\sum}$ in Equation 2 is the summation and average, respectively:

$$\mathcal{E}[b] = \frac{1}{\Delta b} \sum_{k_- \leq |\mathbf{k}| < k_+} E[\mathbf{m}] (\Delta k)^D,$$

236

$$\bar{E}[b] = \frac{1}{K_D[b]} \sum_{k_- \leq |\mathbf{k}| < k_+} E[\mathbf{m}] (\Delta k)^D.$$

237 where $K_D[b]$ is the number of pixels within the bin-range (thus defining the function that
238 averages) and we apply an additional normalization by the bin-width Δb^{-1} (which ensures
239 unit consistency and that $\sum \mathcal{E}[b] \Delta b$ is the total energy). Their relation is found via: $\mathcal{E}[b] =$
240 $K_D[b] \bar{E}[b] / \Delta b$ (assuming the remaining binning properties are kept the same). The bin-shell

¹In the case of 1D data, there are exactly 2 values for each $|\mathbf{k}|$: the positive and negative value.

241 volume coefficients for centered ($b_1 = b - \Delta b/2$, $b_2 = b + \Delta b/2$), or left-edge bins ($b_1 = b$,
242 $b_2 = b + \Delta b$) are:

$$K_D[b] = \frac{V(b_1) - V(b_2)}{(\Delta k)^D}$$

243 where

$$V(r) = \frac{r^D \pi^{D/2}}{\Gamma(D/2 + 1)}$$

244 is the volume of a D -dimensional hypersphere.

245 Future Work

246 It is intended for future development of kea to implement a general interpretation of spectral
247 estimation using arbitrary filters along with automatic implementation of debiasing using the
248 non-parametric local power-law approximation that is currently only available for the ESF
249 method.

250 Another area of potential improvement will be the implementation of synthetic fields with
251 anisotropy. This would be useful to test the statistics of projected fluctuations in more complex
252 scenarios.

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